CECH COHOMOLOGY OF AN INJECTIVE PRESHEAF VANISHES IN POSITIVE DEGREES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The check accent on Cech cohomology is reproduced as the author's own \check H notation.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Allowed standard prerequisites (editorial).

- Cech complex and Cech cohomology of a family in a presheaf of abelian groups. Same chapter, Definition environment labelled definition-cech-complex (source line 1477, Stacks tag 03OL); the statement picks up its notation and assumptions. Supplies the complex Z^bullet_U and the groups H^p-check; the vanishing argument is the counted content.
- The complex Z^bullet_U is exact in negative degrees and Hom(-, I) into an injective is exact. Same chapter, the immediately preceding computation of the complex Z^bullet_U (the author's phrase 'we have just seen'), together with the defining property of an injective object. Supplies the two facts combined in the proof; the combination producing exactness in positive degrees is retained in the counted span.

Source reference. Original definition 03OL: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1477–1477. Same-chapter pointer named in the first line of the counted statement (line 1703) as the source of notation and assumptions; the identification of the Cech complex as a Hom complex and the exactness argument are retained verbatim in the counted span.

Lemma 0.1. *Notation and assumptions as in Definition 03OL. If $\mathcal{I}$ is an injective object of $PAb(\mathcal{C})$, then $\check{H}^p(\mathcal{U}, \mathcal{I}) = 0$ for all $p > 0$.*

Proof. The Čech complex is the result of applying the functor $\mathrm{Hom}_{PAb(\mathcal{C})}(-, \mathcal{I})$ to the complex $\mathbf{Z}^\bullet_{\mathcal{U}}$, i.e.,

$$\check{H}^p(\mathcal{U}, \mathcal{I}) = H^p(\mathrm{Hom}_{PAb(\mathcal{C})}(\mathbf{Z}^\bullet_{\mathcal{U}}, \mathcal{I})).$$

But we have just seen that $\mathbf{Z}^\bullet_{\mathcal{U}}$ is exact in negative degrees, and the functor $\mathrm{Hom}_{PAb(\mathcal{C})}(-, \mathcal{I})$ is exact, hence $\mathrm{Hom}_{PAb(\mathcal{C})}(\mathbf{Z}^\bullet_{\mathcal{U}}, \mathcal{I})$ is exact in positive degrees. □

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.